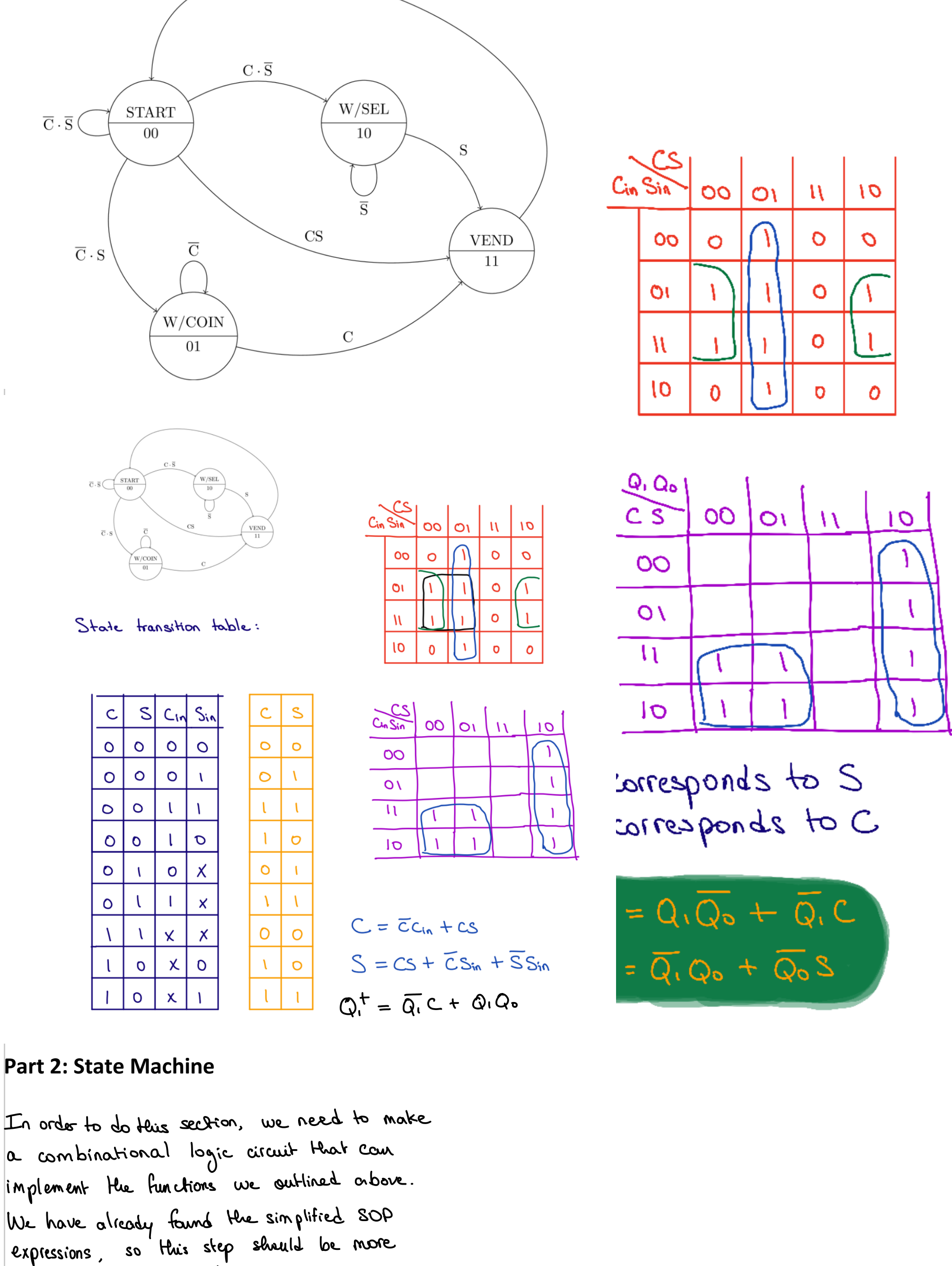


DECA 2.2 - State machines

Sunday, December 11, 2022 4:02 PM

Part 1: Before the lab

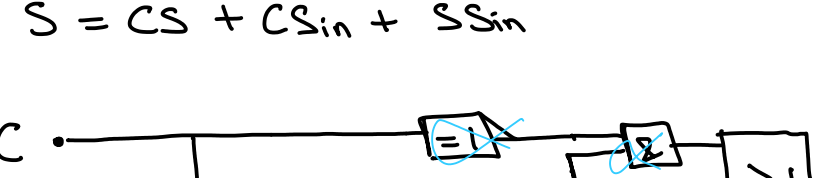


Part 2: State Machine

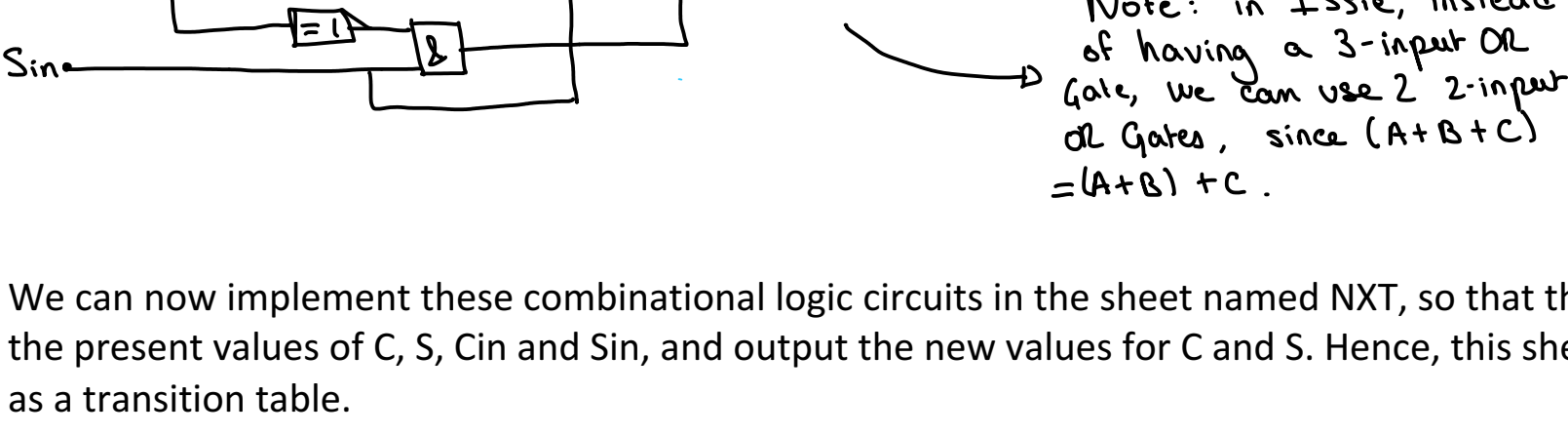
In order to do this section, we need to make a combinational logic circuit that can implement the functions we outlined above.

We have already found the simplified SOP expressions, so this step should be more or less straightforward.

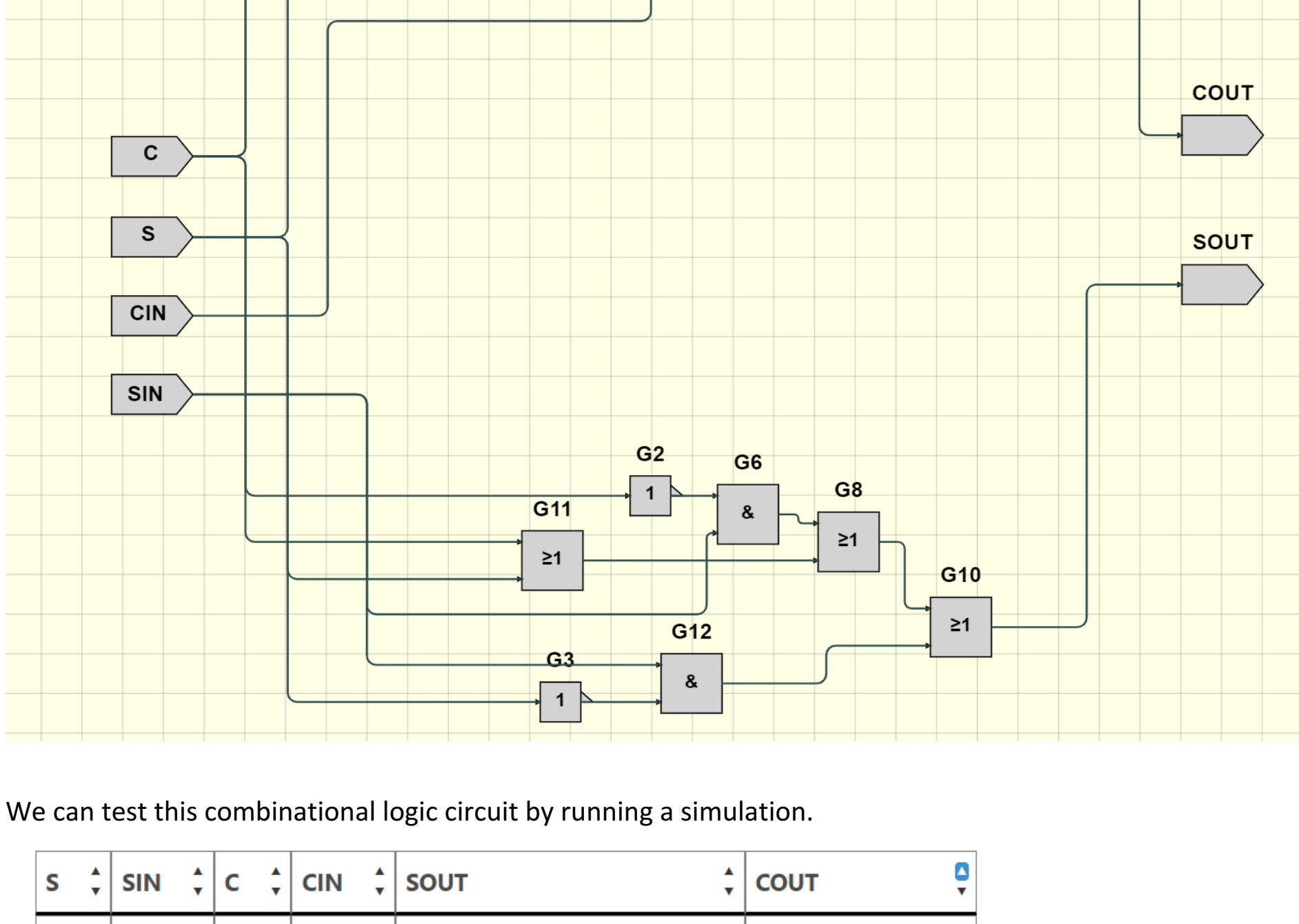
$$C = \bar{C}C_{in} + CS$$



$$S = CS + \bar{C}S_{in} + \bar{S}S_{in}$$



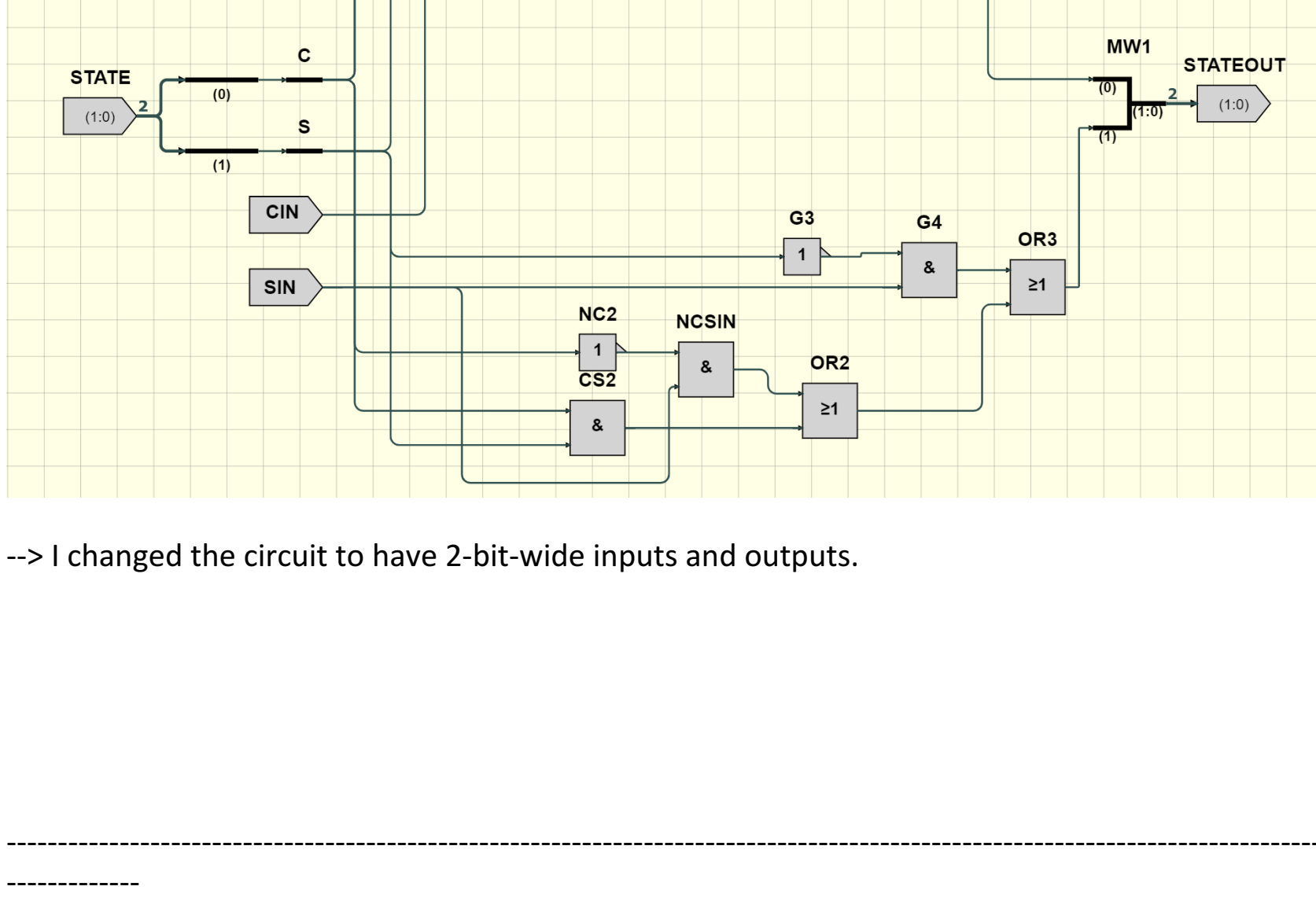
We can now implement these combinational logic circuits in the sheet named NXT, so that this sheet will take in the present values of C, S, Cin and Sin, and output the new values for C and S. Hence, this sheet acts essentially as a transition table.



We can test this combinational logic circuit by running a simulation.

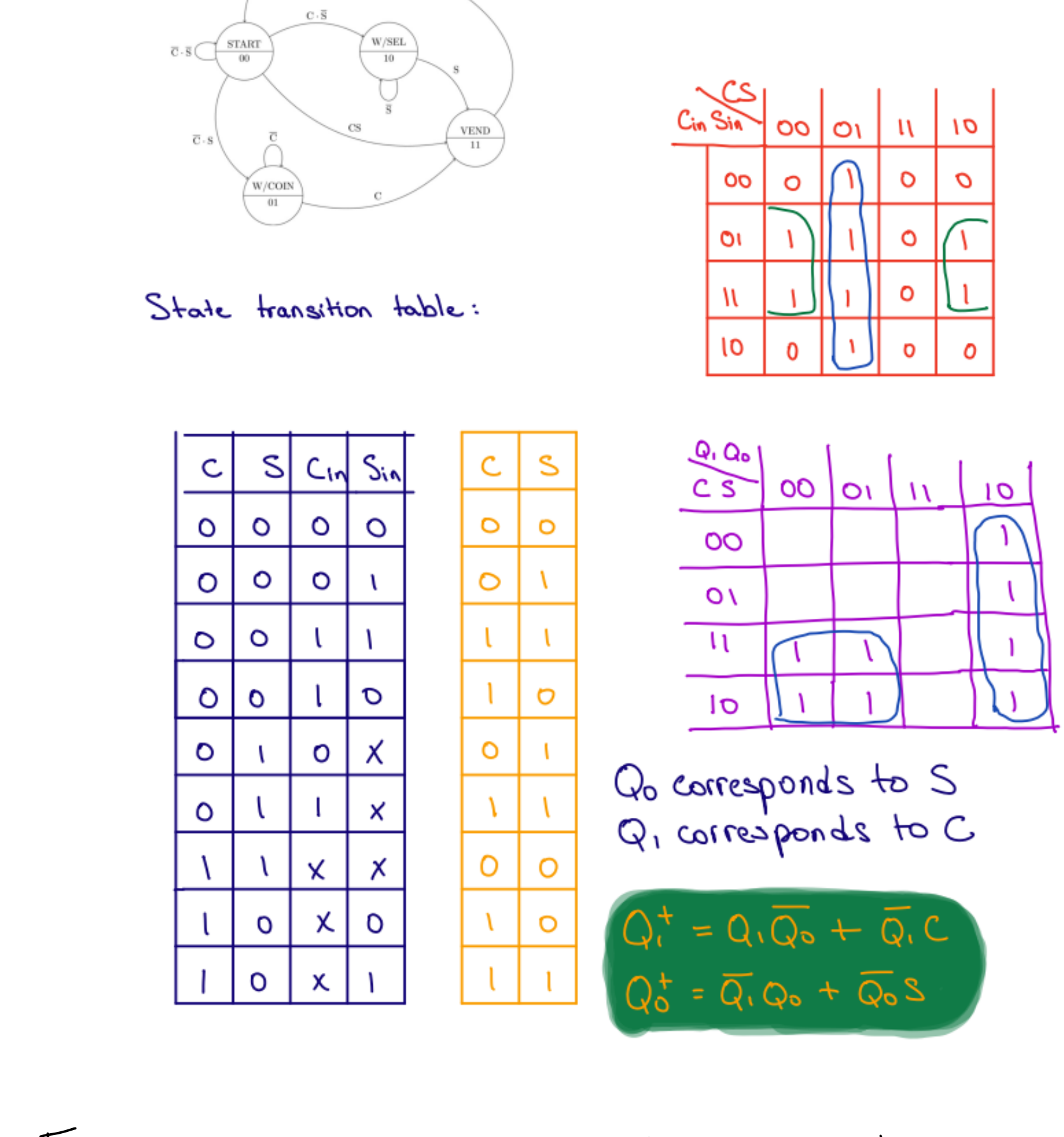
S	SIN	C	CIN	SOUT	COUT
S	SIN	C	CIN	$((\sim C) \& SIN) \vee (C \& S)$	$((\sim C) \& CIN) \vee (C \& S)$

This time, instead of comparing truth tables, I compared algebraic expressions, and confirmed that the circuit worked as we expected.

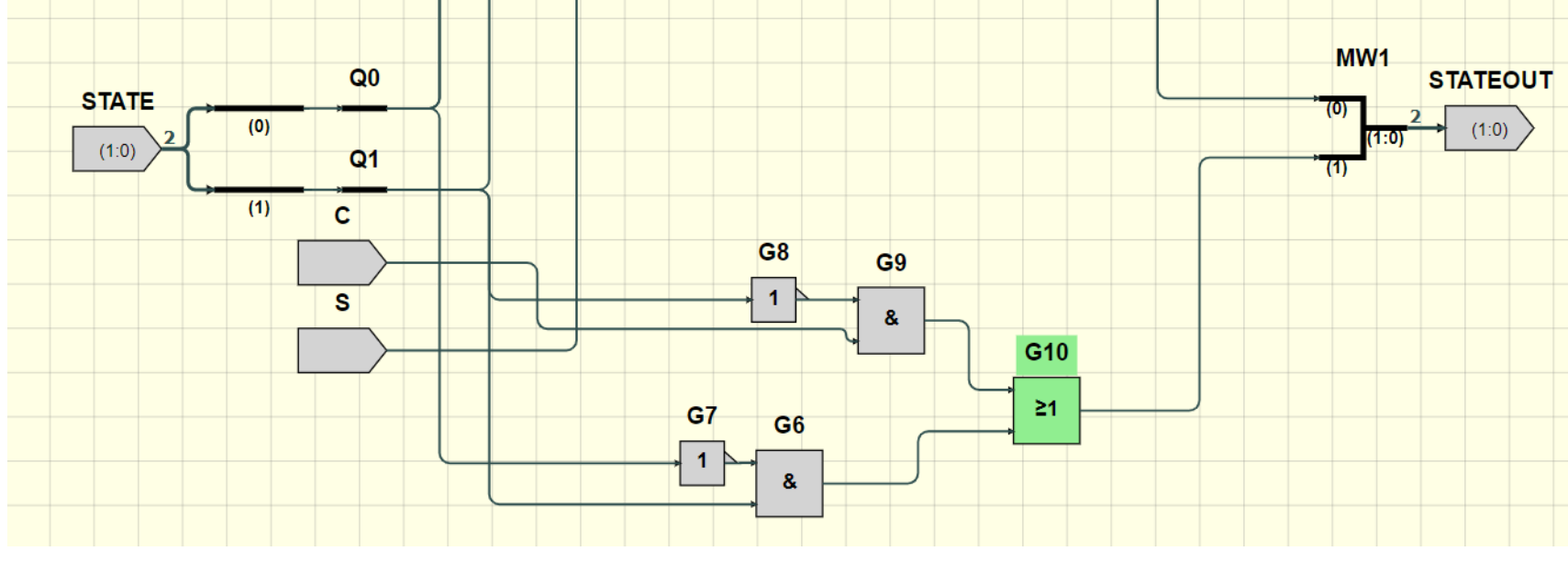
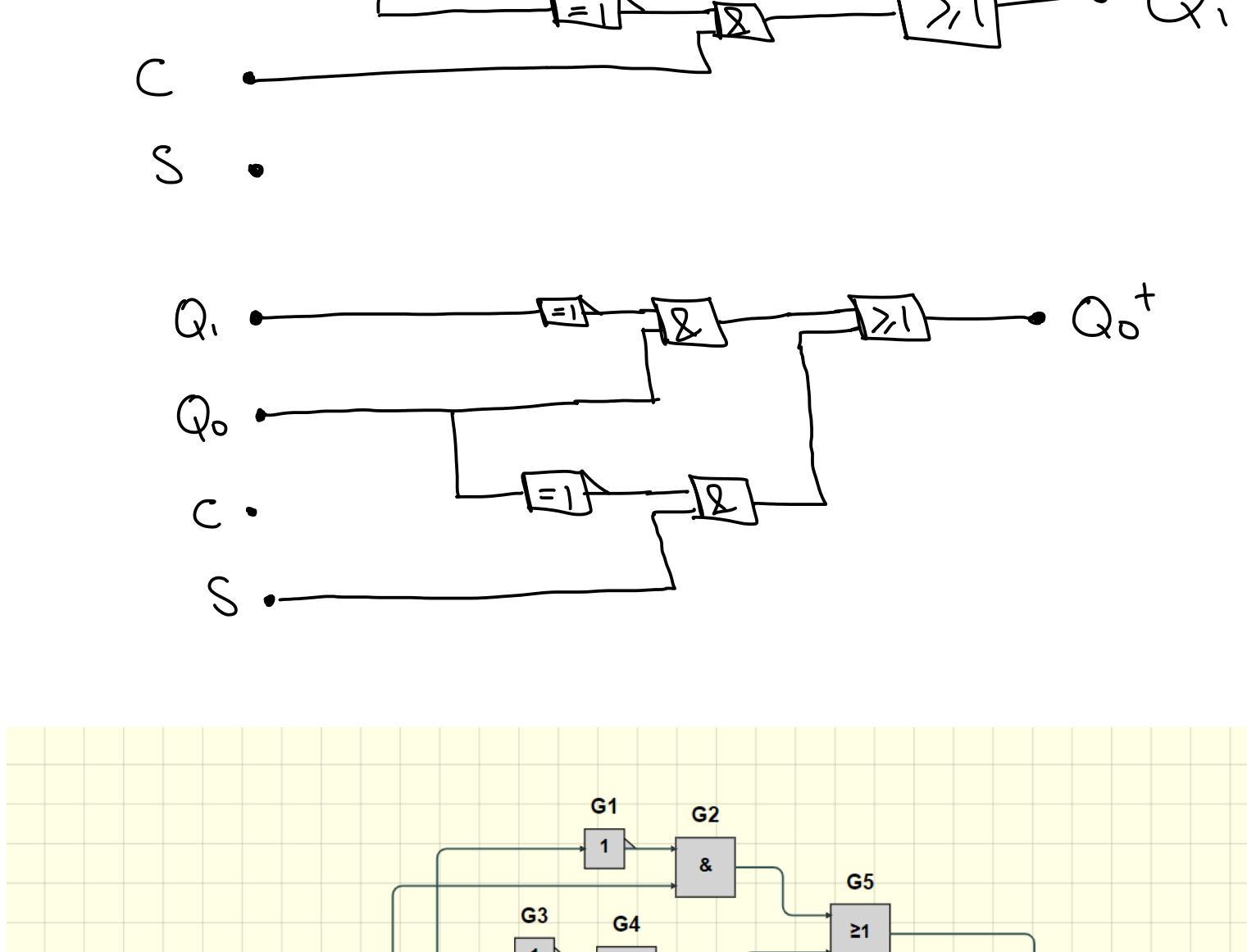


--> I changed the circuit to have 2-bit-wide inputs and outputs.

At this stage I realised that I had mislabelled my Karnaugh maps from the pre-lab section, so I am starting over...



I can now remake my combinational logic circuits:



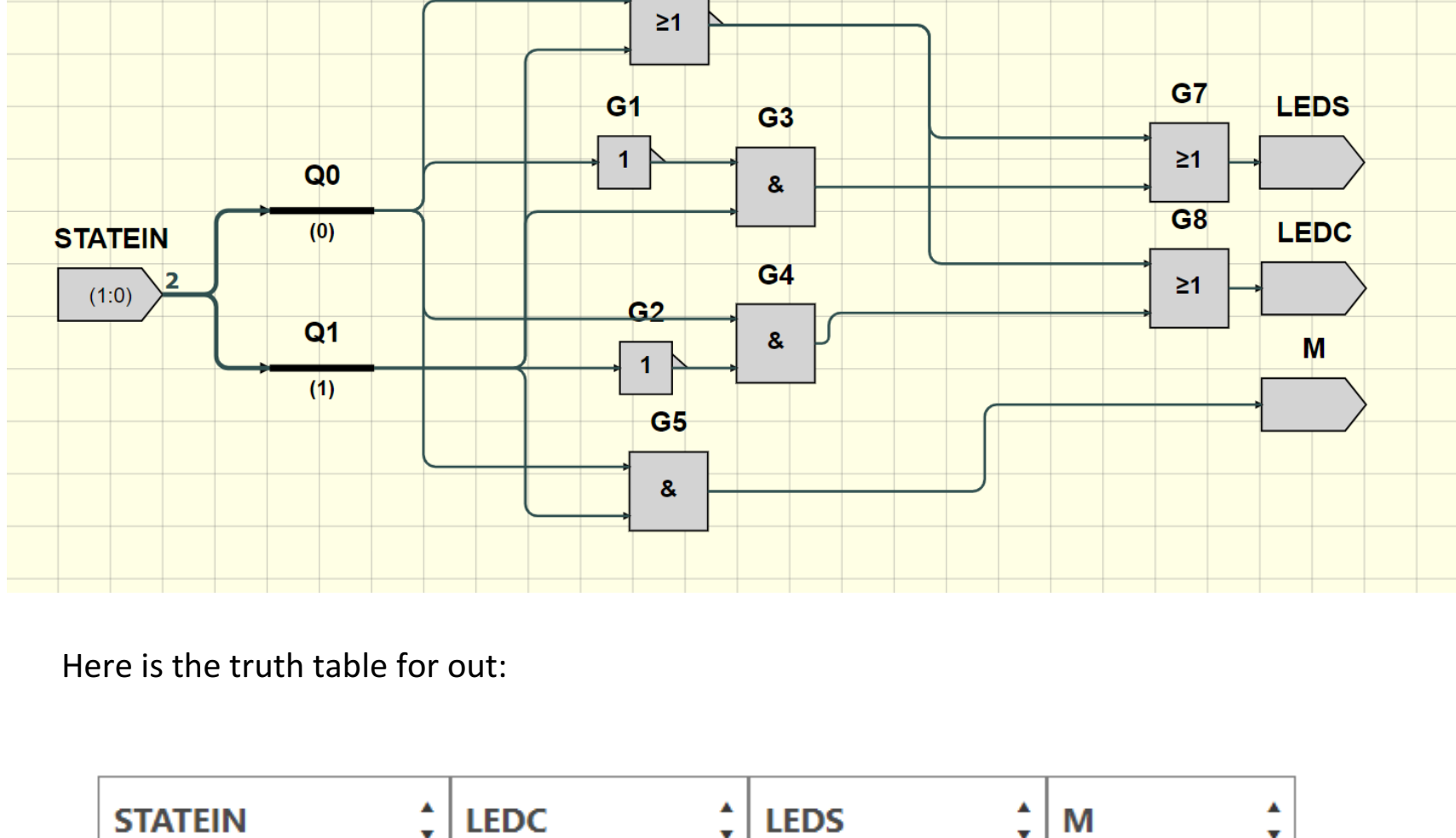
The vend sheet circuit now works correctly.

We now want to make combinational logic "out" to determine what the vending machine should do based on its state, according to the following behaviours:

State	M	LEDS	LEDC
START	0	1	1
W/SEL	0	1	0
W/COIN	0	0	1
VEND	1	0	0

Start $\Leftrightarrow (0, 0) \Rightarrow \bar{Q}_1 \bar{Q}_0$
W/SEL $\Leftrightarrow (1, 0) \Rightarrow Q_1 \bar{Q}_0$
W/COIN $\Leftrightarrow (0, 1) \Rightarrow \bar{Q}_1 Q_0$
VEND $\Leftrightarrow (1, 1) \Rightarrow Q_1 Q_0$

We can implement this with combinational logic with 2-bit-wide input STATE, and 3 1-bit-wide outputs M, LEDS and LEDC.



Here is the truth table for out:

STATEIN	LEDC	LEDS	M
0x0	1	1	0
0x2	0	1	0
0x1	1	0	0
0x3	0	0	1

This is what we expected:

- if STATEIN = 0, then $Q_1 = 0$ and $Q_0 = 0$
 $\Rightarrow C = 0$ and $S = 0$, which means both LEDs should be on, and the motor should be off.
- if STATEIN = 2, then $Q_1 = 1$ and $Q_0 = 0$
 $\Rightarrow C = 1$, $S = 0$, which means LEDS should be on, and this is indeed what we have. Additionally, motor is in fact off.
- if STATEIN = 3, then $Q_1 = 1$ and $Q_0 = 1$,
 $\Rightarrow C = 1$, $S = 1$, so both LEDs should be off and the motor should be on, which is indeed what we get!

